MILITARY SPECIFICATION

CANNON, 155MM HOWITZER: M185

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cannon, 155MM Howitzer: M185 consisting of a tube, breech mechanism, bore evacuator, and muzzle brake.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannon, Mortars, Recoilless Rifles and Recoilless Guns: General Specification for

MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-LCB-SAS, Watervliet, N.Y. 12189-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1189 - Standard Symbology for Marking Unit Packs, Outer Containers, and Selected Documents

MIL-STD-1235 - Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U. S. Army Armament, Munitions and Chemical Command

F 11578365 - General Drawing, Cannon, 155MM Howitzer: M185

Master LQAP 11578365 - Master Index of Quality Assurance Provisions for Cannon, 155MM Howitzer: M185

P11578365 - Packaging Data Sheet, Cannon, 155MM Howitzer: M185

QAP-APPENDIX WVA - General Quality Assurance Provisions

PUBLICATION

DA Form 2408-4 - Weapon Record Data

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 -Standard Practice for Commerical Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, drawing requirements shall have first order of precedence, followed by the text of this specification, then the other referenced specifications and standards.

3. REQUIREMENTS

3.1 First article. The contractor shall submit a first article of cannon, components thereof, or both unless it is specifically waived in the contract (see 6.2.1). No first article requirements shall be waived without review by the responsible engineering activity and approval by the responsible product assurance activity.

3.2 General requirements. Cannon and parts thereof shall conform to the General Drawing F11578365, the requirements of this specification, and MIL-C-13931.

3.2.1 Marking. When specified, marking shall be in accordance with the requirement specified in MIL-C-13931.

3.3 Functioning. All parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. Camming of the operating crank during counter-recoil or rotating the operating handle rearward and to the right shall force the breechblock operating rack downward and to the right. This action shall cause the breechblock to rotate counterclockwise to disengage the sectored threads. At this time sufficient force shall be exerted, through action of the operating crank, to overcome the carrier-latching plungers thereby releasing the plungers from the catches attached to the breech ring and in turn allowing the carrier and breechblock assembly to swing to the open position. The breech mechanism shall be retained in the open position by means external to the cannon, i.e., engagement of the operating crank roller in the operating cam of the mount.

3.3.2 Breech closing. With the adjustor set on the minimum preload position and the cannon at the normal loading elevation, closing of the breech shall be accomplished by lifting the

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operating-cam of the mount clear of the operating crank rollers. Release of the crank shall allow the torsion springs to drive the crank clockwise and forcibly return the breechblock and carrier to the closed position. Action of the rack springs against the breechblock operating rack shall rotate the breechblock clockwise to engage the sectored threads. If the breech fails to close fully, manual closing shall be accomplished by engaging the clutch in the notch of the operating crank assembly and rotating the operating handle to the left and forward. The operating handle shall latch in the closed position. Counter-clockwise rotation of the adjustor shall increase the pre-load on the torsion springs.

3.3.3 Obturator surface. The obturator surfaces shall conform to the requirements prescribed in MIL-C-13931.

3.3.4 Block assembly. The block assembly, with the specified percussion-type firing mechanism mounted in position, shall slide in the slot in the housing assembly into the extract position when the breechblock is moved to the open position. As the extract position is reached, the extractor shall forcibly eject the primer out of the spindle chamber. When the breechblock is moved to close the breech, the block assembly shall be automatically cammed from the extract to the load position while the extractor is cammed forward against the rear of the spindle. The block assembly shall be capable of moving to the closed position only when the breechblock is closed and locked. In this position the block assembly shall also be capable of being moved manually into the load and extract positions.

3.4 Primer headspace. Primer headspace, the distance between the front face of the block assembly and the rear face of the obturator spindle, shall be not less than 0.054 inch nor more than 0.062 inch.

3.5 Firing pin protrusion and retraction. The firing pin protrusion measured from the front face of the block shall be not less than 0.114 inch and not more than 0.118 inch in the fired position. Firing pin retraction behind the front face of the block shall be not less than 0.008 inch.

3.6 Performance.

3.6.1 High pressure resistance. The cannon and tubes shall be capable of withstanding a chamber pressure of 43,500 $\pm$ 1500 psi.

3.6.2 Material soundness. After completion of acceptance testing, all parts shall be free of cracks and fractures and shall comply with soundness requirements when specified on the applicable drawings and QAPs.

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3.6.3 Bore enlargement. Bore diameters across lands, after firing no more than 15 rounds, shall not exceed the before firing diameters by more than 0.002.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 The inspections set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of his responsibility for assuring that all supplies and services submitted to the Government for acceptance conform to all requirements of the contract.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. First Article-Initial Production Test, FA-IPT, (see 4.5).

c. Quality conformance inspection, (see 4.6).

4.4 First Article inspection. Unless otherwise specified in the contract, (see 6.2.1), First Article cannon and specified parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 11578365. First Article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First Article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

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4.5 First Article - Initial Production Test (FA-IPT). When specified in the contract, (see 6.2.1), First Article cannon shall be subjected to Initial Production Testing by the Government or any agency specified by the Government and may be conducted at a Government facility to assure compliance with the requirements of this specification and any additional requirements listed in the contract. Initial Production Testing procedures and the sample size of first article cannon shall be as specified in the contract. First article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.6 Quality conformance inspection. Quality conformance inspection is listed in Table I.

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TABLE I. Quality conformance inspection.

Categories/Characteristics	Requirement Paragraph	Inspection/Test Method Paragraph
Examinations		
Primer headspace	3.4	4.6.4.1
Firing pin protrusion and retraction	3.5	4.6.4.2
Material soundness	3.6.2	4.6.4.3
Packaging	5.	4.6.4.4
Functional inspections		
Cannon	3.3	4.6.5.1
Breech mechanism	3.3.1	4.6.5.2
Breech closing	3.3.2	4.6.5.2
Block assembly	3.3.4	4.6.5.3
Tests		
Proof acceptance	3.6	4.6.6.1
Proof firing		4.7.1
Proof pressure simulation		4.7.2

4.6.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance with the requirements of this specification, MIL-C-13931, QAPs listed in Master LOAP 11578365, and QAP-APPENDIX-WVA.

4.6.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.6.3 Inspection equipment. Inspection equipment used in the inspection of parts, subassemblies, and assemblies is prescribed in applicable item QAPs.

4.6.3.1 Acquisition, maintenance, and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquisition,

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maintenance, and disposition of inspection equipment prescribed in applicable item QAPs, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, or the contract, shall be in accordance with MIL-I-45607.

4.6.3.2 Accuracy of standard measuring equipment. When commercial and modified-commercial inspection equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6.4 Examinations.

4.6.4.1 Primer headspace. Primer headspace shall be gaged on each cannon with the applicable gage prescribed in the applicable item QAP, or with an approved equivalent gage. Cannons failing to meet the requirements of 3.4 shall be subject to rejection.

4.6.4.2 Firing pin protrusion and retraction. Firing pin protrusion and retraction shall be gaged on each cannon with the applicable gages prescribed in the applicable item QAP, or with approved equivalent gages. Cannons failing to meet the requirements of 3.5 shall be subject to rejection.

4.6.4.3 Material soundness. The outer surfaces of each tube, all surfaces of each breechblock, breech ring, carrier, evacuator, and muzzle brake shall be magnetic particle tested in accordance with applicable drawings and specifications. The inner surface of the tube shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejection.

4.6.4.4 Packaging inspection. Packaging inspection for the level designated in the contract shall be performed in accordance with the applicable packaging data sheets and MIL-P-14232. Packages failing to comply with the packaging requirements shall be subject to rejection.

4.6.5 Functional inspections.

4.6.5.1 Cannon. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each cannon.

4.6.5.2 Breech mechanism. Each breech mechanism shall be operated manually to determine smoothness of operation. Action of the operating handle shall be observed to insure that the handle will latch in the closed position after each complete cycle of operation. Any evidence of interference, erratic move-

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ment, or malfunction of any parts shall be cause for rejection.

4.6.5.3 Block assembly. Each block assembly shall be operated manually to determine whether the requirements are met. Any deviation from the requirements of 3.3.4 shall be cause for rejection.

4.6.6 Tests.

4.6.6.1 Proof acceptance. Cannon proof acceptance program shall consist of proof firing, proof pressure simulation tests, or both. Cannon and replacement tubes to be proof fired shall be fired using the test method of 4.7.1. Failure to comply with the requirements of 3.6 shall be cause for rejection. Cannon breech mechanisms to be simulation proof tested shall be proof tested using the test method of 4.7.2. Failure to comply with the requirements of 3.6.1 and 3.6.2, shall be cause for rejection. Proof firing shall be done on a sampling basis and shall be conducted as follows:

a. Using plan CSP-1 of MIL-STD-1235 and an AQL of 1.0, the Product Assurance Directorate at Watervliet Arsenal shall designate the appropriate initial quantity (i) and sample frequency (f), in the contract (see 6.2.1).

b. Tne first "i" quantity of cannon or tubes produced in a particular buy shall be proof fired. When the "i" quantity has been satisfied, sampling shall be put into effect using the "f" sampling frequency. All cannon not proof fired shall have the breech mechanisms simulation proof tested in the Product Assurance weapon test facility at Watervliet Arsenal or other Government approved test facility.

c. Successive production buys shall not require "i" quantities. However, if the time lag between two successive buys exceeds one year, or if a new cannon or tube contractor is utilized, a new "i" quantity shall be proof fired.

d. Proof fired cannon and replacement tubes, and simulation proof tested breech mechanisms shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data) as specified by Product Assurance Directorate at Watervliet Arsenal.

e. When specified in the contract(see 6.2.1), the weapon simulation test facility shall be qualified as follows: a sample of breech mechanisms of cannon to be sent to the proving ground for proof firing shall be subjected initially to simulation proof testing at the weapon simulation test facility.

4.7 Test methods.

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4.7.1 Proof firing. Proof firing shall be conducted at a Government approved test site. The procedure shall be as follows:

a. Each cannon or replacement tube shall be proof fired within the proof pressure range specified in 3.6.1. If the lower limit is not attained, additional rounds shall be fired until the minimum value is attained. The number of additional rounds shall not exceed five.

b. A facility mount may be used if necessary.

c. Chamber pressure shall be measured and recorded.

d. After proof firing:

(1) The bore of each tube shall be examined for bore enlargement using an approved gage (see 3.6.3.).

(2) Each tube shall be borescoped and magnetic particle inspected (see 4.6.4.3).

(3) The evacuator and muzzle brake of each tube assembly shall be magnetic particle inspected (see 4.6.4.3).

(4) Each breech mechanism shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.6.5).

(5) Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be examined to assure freedom from cracks and fractures. Each breech ring and breechblock and carrier shall be magnetic particle inspected (see 4.6.4.3).

e. Cannon found to be defective at the proving ground shall be reported by teletype to the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189-5000 and Benet Weapons Lab, Watervliet Arsenal, Watervliet, NY 12189-5000 within 12 hours followed by an Equipment Performance Report (EPR), for critical incidents. If the incident is not critical, an EPR is requird within ten working days.

4.7.2 Proof pressure simulation. All cannon not proof fired at a proving ground shall have the breech mechanism proof tested in accordance with the following procedure:

a. Each breech mechanism shall be tested by means of suitable hydraulic pressure testing equipment at a pressure within the proof pressure range specified in 3.6.1.

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b. Cycle time to achieve test pressure and release shall not exceed three minutes. Test pressure shall be immediately released upon its attainment with dwell time at test pressure not to exceed three seconds.

(1) When difficulty is encountered in attaining test pressure, the number of attempts in which 10 percent or less of the test pressure is attained, shall not be counted. The number of attempts in which over 10 percent of the test pressure is attained, shall not exceed five.

(2) If the cause of difficulty is not identified and corrected within five pressure cycles, the breech mechanism shall be subjected to proof firing as part a cannon (see 4.7.1). A notification of this condition shall be submitted with the breech mechanism to the proving ground.

c. The peak pressure shall be recorded for each test and attached to inspection records identifiable with the specific breech mechanism undergoing test.

d. After the pressure test, each breech mechanism shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.6.5.2 and 4.6.5.3).

e. Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. Each breech ring, breechblock, and carrier shall be magnetic particle inspected (see 4.6.4.3).

5. PACKAGING

5.1 Cannon - Levels A, B and C. Preservation, packaging and packing of the cannon shall be in accordance with Packaging Data Sheet P11578365 for the level of protection specified in the contract (see 6.2.1).

5.2 Repair parts.

5.2.1 Levels A, B and C. Repair parts preservation, packaging and packing shall be in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2.1).

5.2.2 Level X-industrial packaging. Repair parts preservation, packaging, and packing shall be in accordance with ASTM D 3951 when Level X protection is specified in the contract (see 6.2.1).

5.3 Marking.

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5.3.1 Level A, B and C. Marking of unit packages and exterior shipping containers shall be in accordance with Packaging Data Sheet P11578365 for the level of protection specified in the contract (see 6.2.1).

5.3.2 Level X - Industrial. Marking shall be in accordance with ASTM D 3951.

5.3.3 Bar Code Marking. When specified, (see 6.2.1.n) bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The cannon specified herein is intended to be used in the Howitzer, Medium, Self-Propelled, 155MM M109A1, M109A2, and M109A3 Systems.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.

b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.1).

c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

d. All inspection and test data that is required.

e. At whose expense a retest may be performed.

f. Serial numbers for the cannon and tube assemblies.

g. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.6.3).

h. Availability of inspection equipment from the Government.

i. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment

j. Applicable acceptance test procedures, initial consecutive quantity (i), and sampling frequency (f) (see 4.6.6).

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k. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

l. When applicable, the sampling plan for the actual firing of tubes.

m. The levels of preservation, packaging, and packing required for cannon (see 5.1) and repair parts (see 5.2).

n. When applicable, bar code marking requirements (see 5.3).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2.2.1 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.1	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer).

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-STD-45662.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing activity:

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Project No. 1025 A 157

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-45964A	CANNON, 155MM HOWITZER: M185

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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